

Moller scattering 2

For transition amplitudes M_t we have

$$M_t = \iint \underbrace{\bar{\psi}_4(y)(-ie\gamma^\mu)\psi_2(y)}_{\text{vertex}} \underbrace{G(x,y)}_{\substack{\text{photon} \\ \text{propagator}}} \underbrace{\bar{\psi}_3(x)(-ie\gamma_\mu)\psi_1(x)}_{\text{vertex}} d^4x d^4y$$

where $G(x, y)$ is the photon propagator

$$G(x, y) = \frac{1}{(2\pi)^4} \int \frac{\exp(-ip(x-y))}{p^2 + i\epsilon} d^4p$$

For Moller scattering the normalized wave functions are

$$\begin{aligned}\psi_1(x^\mu) &= \sqrt{2EV} u_1 \exp(-ip_1^\mu g_{\mu\nu} x^\nu) \\ \psi_2(x^\mu) &= \sqrt{2EV} u_2 \exp(-ip_2^\mu g_{\mu\nu} x^\nu) \\ \psi_3(x^\mu) &= \sqrt{2EV} u_3 \exp(-ip_3^\mu g_{\mu\nu} x^\nu) \\ \psi_4(x^\mu) &= \sqrt{2EV} u_4 \exp(-ip_4^\mu g_{\mu\nu} x^\nu)\end{aligned}$$

For clarity we abbreviate as

$$\begin{aligned}\psi_1(x) &= \sqrt{2EV} u_1 \exp(-ip_1 x) \\ \psi_2(x) &= \sqrt{2EV} u_2 \exp(-ip_2 x) \\ \psi_3(x) &= \sqrt{2EV} u_3 \exp(-ip_3 x) \\ \psi_4(x) &= \sqrt{2EV} u_4 \exp(-ip_4 x)\end{aligned}$$

Substitute for the wave functions in M_t and factor the spinors.

$$\begin{aligned}M_t &= -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \\ &\quad \times \iint \exp(ip_4 y) \exp(-ip_2 y) G(x, y) \exp(ip_3 x) \exp(-ip_1 x) d^4x d^4y\end{aligned}$$

Substitute for $G(x, y)$.

$$\begin{aligned}M_t &= -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \\ &\quad \times \frac{1}{(2\pi)^4} \iiint \exp(ip_4 y) \exp(-ip_2 y) \frac{\exp(-ip(x-y))}{p^2 + i\epsilon} \exp(ip_3 x) \exp(-ip_1 x) d^4x d^4y d^4p\end{aligned}$$

Combine exponentials.

$$\begin{aligned}M_t &= -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \\ &\quad \times \frac{1}{(2\pi)^4} \iiint \frac{\exp(i(p_4 - p_2 + p)y) \exp(i(p_3 - p_1 - p)x)}{p^2 + i\epsilon} d^4x d^4y d^4p\end{aligned}$$

The integrals over x and y are solved as

$$\begin{aligned}\int \exp(i(p_3 - p_1 - p)x) d^4x &= (2\pi)^4 \delta^4(p_3 - p_1 - p) \\ \int \exp(i(p_4 - p_2 + p)y) d^4y &= (2\pi)^4 \delta^4(p_4 - p_2 + p)\end{aligned}$$

Hence

$$M_t = -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) (2\pi)^4 \int \frac{\delta^4(p_4 - p_2 + p) \delta^4(p_3 - p_1 - p)}{p^2 + i\epsilon} d^4p$$

The integral over p vanishes except for

$$p = p_3 - p_1 = p_2 - p_4$$

Hence

$$M_t = -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) (2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_3 - p_1)^2}$$